Predicting High-Yield Credit Spreads: A Comparative Analysis of LASSO Regression and XGBoost Environments

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1. Problem Identification

1.1 Research Question

The central objective of this study is to determine to what extent macroeconomic indicators and financial market sentiment can predict daily changes in U.S. High Yield (HY) Credit Spreads. Specifically, this research evaluates whether credit risk dynamics are driven by simple, proportional linear relationships or by complex, regime-dependent interactions. To test this hypothesis, the study compares the predictive accuracy and structural interpretability of a linear, regularized model (LASSO Regression) against a non-linear, tree-based ensemble approach (XGBoost). By utilizing high-frequency daily data, this research seeks to answer whether the inherent noise of daily market fluctuations obscures macroeconomic signals, or if advanced machine learning algorithms can successfully isolate meaningful, predictive patterns. Ultimately, the question asks: does the high-yield credit market behave as a continuous linear function of macroeconomic indicators, or does it require algorithms capable of capturing asymmetric market shocks and conditional volatility?

1.2 Background Context

Credit spreads—defined as the difference between the yield on corporate bonds and risk-free government Treasuries—serve as a vital "barometer" for the health of the broader economy. For institutional investors and policymakers, these spreads represent the market's collective, real-time assessment of corporate default risk, liquidity stress, and overall risk appetite. Historically, structural credit risk models posit that credit spreads are influenced by a convergence of firm-specific and macroeconomic factors (Merton 1974). This study utilizes several critical features grounded in this economic theory:

- **Equity Market Performance:** As a junior claim on a firm's assets, stock prices (e.g., the S&P 500) provide real-time signals regarding a firm's "distance to default." Positive equity momentum generally indicates rising asset values, which in turn compresses credit spreads (Merton 1974).
- **Volatility and Fear:** The VIX Index captures investor "nervousness" and expected market volatility. Higher volatility increases the probability that firm asset values will fall below

outstanding debt obligations, historically correlating with sharp widenings of risk premiums.

- **Monetary Policy and the Yield Curve:** Changes in the 2-Year Treasury rate reflect the absolute cost of capital and impending refinancing risk for highly leveraged firms. Concurrently, the 10Y-2Y yield curve slope acts as a premier leading economic indicator, reflecting the market's long-term growth and inflation expectations (Estrella and Mishkin 1998).
- **Systemic Financial Conditions:** Liquidity indices capture underlying frictions in the banking system that often precede broader credit contractions and corporate defaults (Gilchrist and Zakrajšek 2012).

While traditional econometrics heavily relies on linear Ordinary Least Squares (OLS) assumptions, credit markets are notoriously prone to "fat tails," heteroskedasticity, and sudden, asymmetric shocks (e.g., the 2008 Global Financial Crisis and the 2020 COVID-19 pandemic liquidity freeze). This project leverages comprehensive datasets from Federal Reserve Economic Data (Federal Reserve Bank of St. Louis 2024) and the Bloomberg Terminal (Bloomberg L.P. 2024) to test if advanced Machine Learning techniques can overcome the limitations of classical linear models when confronted with these erratic, non-normal market behaviors.

1.3 Importance of the Question

Understanding the underlying drivers of high-yield credit spread movements is of paramount importance to both academia and industry practitioners for three primary reasons:

- **Risk Management and Portfolio Optimization:** For institutional fixed-income portfolios, credit spreads are the primary source of excess return (alpha) as well as drawdown risk. Accurate forecasting models allow portfolio managers to implement more effective dynamic hedging strategies, optimize asset allocation across credit tiers, and better calculate Value at Risk (VaR) ahead of severe credit contractions (Duffie and Singleton 2003).
- **Macroeconomic Forecasting and Policy Formulation:** Because corporate credit markets often lead the "real" economy, predicting spread widening can serve as an early warning

signal for broader economic recessions (Gertler and Lown 1999). Policymakers and central banks closely monitor high-yield spreads to gauge the transmission efficiency of monetary policy and to intervene before corporate liquidity crises trigger systemic economic contagion (Gilchrist and Zakrajšek 2012).

- **Model Robustness and the "Black Box" Dilemma:** In the context of "Supervised Learning Pipelines," this research rigorously tests the fundamental trade-off between structural interpretability and predictive power (Hastie, Tibshirani, and Friedman 2009). LASSO regression provides transparent, penalized feature selection, making its outputs highly interpretable and suitable for strict regulatory compliance. Conversely, XGBoost offers superior flexibility in mapping non-linearities and variable interactions but suffers from reduced transparency. Determining which model performs better in a noisy, high-frequency daily time-series context provides valuable insights into the structural efficiency of high-yield bond pricing, guiding quantitative researchers in selecting the appropriate mathematical tools for financial forecasting (Gu, Kelly, and Xiu 2020).

2. Variable Selection and Economic Justification

The model utilizes a supervised learning approach to forecast corporate credit spreads. The selection of independent variables (X) is grounded in structural credit risk theory and empirical macroeconomic research.

2.1 Target Variable

- **High Yield Credit Spread (FRED: BAMLH0A0HYM2) (USHY in the code file):** As the dependent variable (Y), the Option-Adjusted Spread (OAS) on the ICE BofA US High Yield Index is used. This represents the risk premium demanded by investors above the risk-free rate. Following **Collin-Dufresne et al. (2001)**, credit spreads are treated as being driven by a significant common factor related to systemic risk and market liquidity.

2.2 Independent Variables (Features)

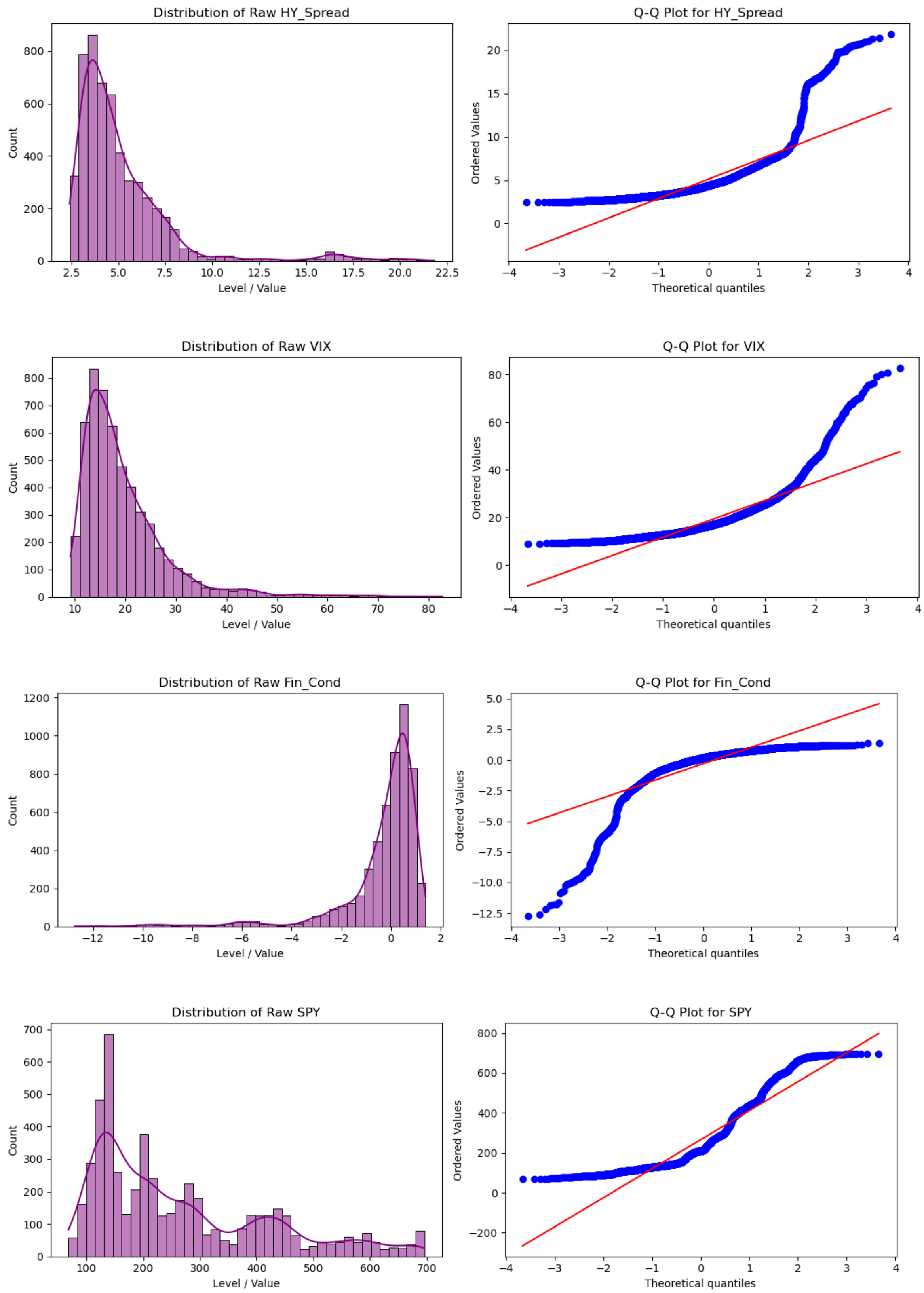
Variable	Metric[1]	Economic Justification & Citation
Market Volatility	VIX Index (VIX)	**Equity Volatility:** Based on Merton's (1974) structural model, an increase in volatility raises the probability that a firm's asset value will hit its "default point." High VIX levels correlate strongly with spread widening.
Equity Returns	S&P 500 (SPY)	**Firm Value Momentum:** Merton (1974) posits that equity acts as a junior claim on firm assets. Declining equity prices signal a reduction in the "distance to default," necessitating higher credit premiums.
Yield Curve Slope	T10Y2Y (yeildslope)	**Macroeconomic Outlook:** As demonstrated by Estrella and Mishkin (1998), the 10Y-2Y spread is a premier leading indicator of recessions. Since default cycles are pro-cyclical, a flattening or inverted curve serves as a critical predictive feature for credit stress.
Interest Rate Level	2-Year Treasury(yeild_2yr)	**Refinancing Risk:** While the slope provides the outlook, the absolute level of the 2-Year yield represents the base cost of capital. Longstaff and Schwartz (1995) argue that the level of the risk-free rate influences the valuation of the "tax shield" and the probability of default.

[1] The variable for each used in the VS code sheet is mentioned in the brackets

Variable	Metric[1]	Economic Justification & Citation
Financial Conditions	BFCIUS (FinCondn)	**Financial Conditions of the banking system:** Following Gilchrist and Zakrajšek (2012), financial conditions indices capture "frictions" in the banking system. Tightening conditions often precede credit contractions, making this a vital feature for non-linear ML models.
Industrial Production	INDPRO (indpro)	**Real Economic Activity/Macroeconomic growth:** This provides a direct measure of physical output. Declines in industrial production signal a weakening "real economy," which increases the fundamental credit risk of high-yield issuers.
Inflation Expectations	10-Year Breakeven Inflation Rate (infexpect)	**Growth and Pricing Power:** Rising inflation expectations often signal economic expansion and increased corporate pricing power, which typically leads to spread compression (Merton 1974).

Daily data were collected from Bloomberg and FRED to capture the high-frequency dynamics of credit risk, totaling 5,365 observations after cleaning. The dataset spans a 20-year period, from March 1, 2006, to March 13, 2026. For the analysis, two distinct datasets were constructed: a primary daily-frequency database and a secondary monthly database created by resampling the daily observations. The resulting monthly dataset comprises 240 observations.

2.3 Data Transformation and Stationarity


Distribution of Raw HY_Spread
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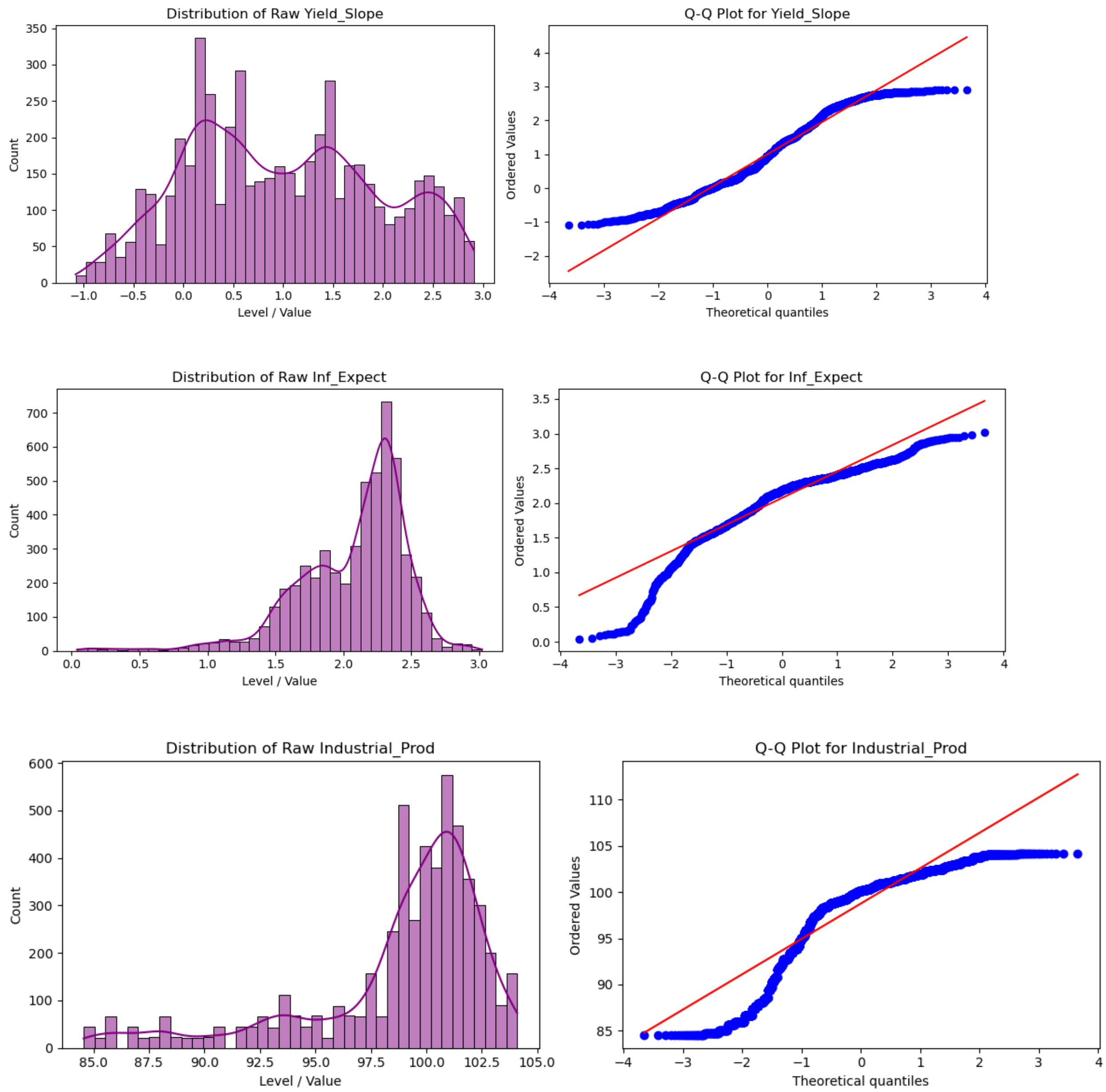

Fig 1: Distribution and QQ Plots for our dependent and independent variables.

An exploratory visual analysis of the raw financial variables revealed significant deviations from normality, as evidenced by heavily skewed histograms and a pronounced curvature away from the theoretical normal distribution in the Q-Q plots. These visual diagnostics strongly indicated that the raw index levels and interest rates contained underlying trends, making the data inherently non-stationary over the sampled period.

To ensure the integrity of the machine learning models, these inputs had to be transformed. Consequently, all non-stationary price-level data (e.g., SPY) were transformed into log-returns, while credit spreads and interest rates were maintained in levels or first differences to preserve the signal of the risk premium. This crucial preprocessing step mitigates the risk of spurious regressions, a common pitfall in financial time-series modeling.

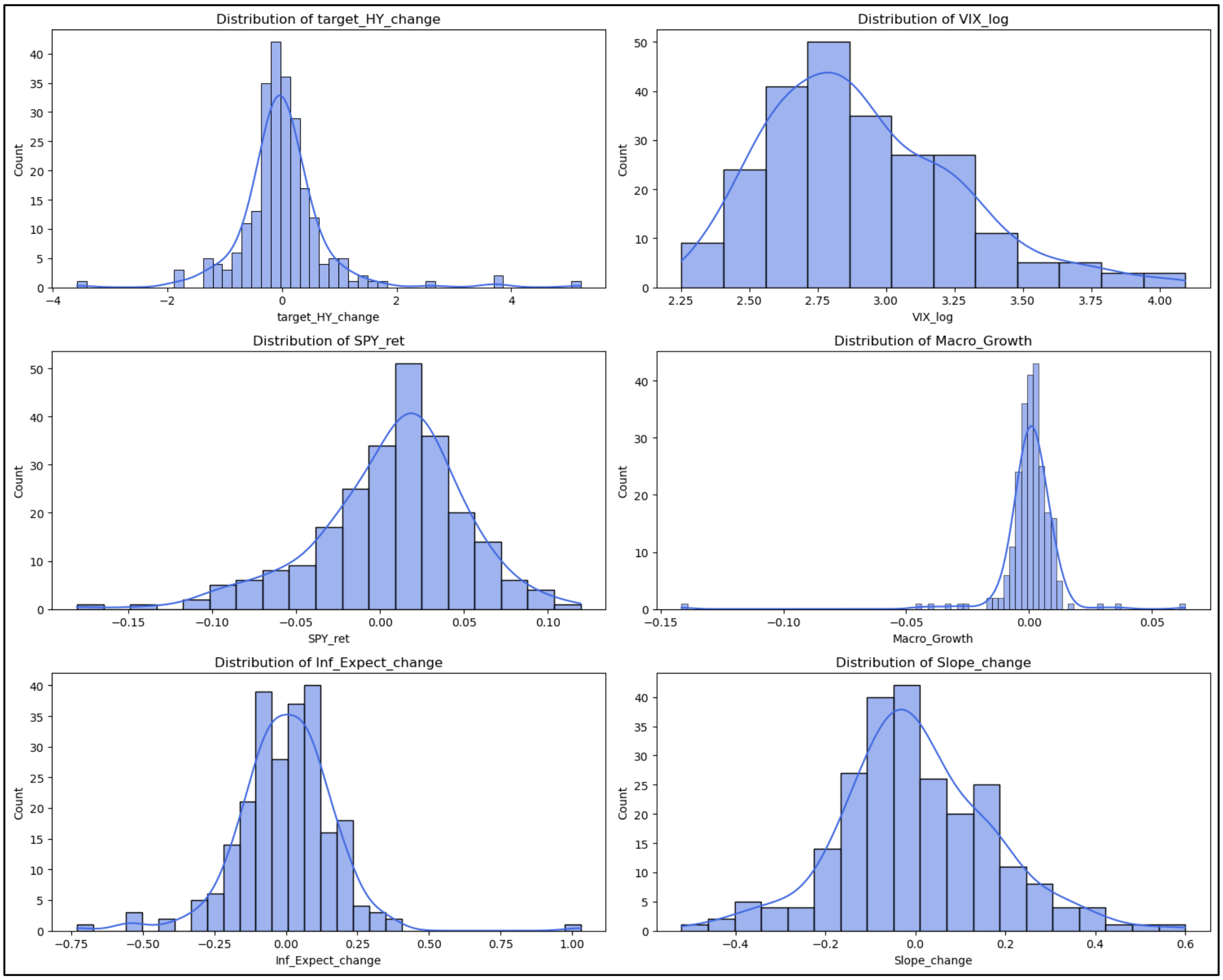

Fig 2: Data post transformation.

3. Analysis and Insights

3.1 The Monthly Model: Baseline Performance

Initially, the study was conducted at a monthly frequency to accommodate the inclusion of Industrial Production (INDPRO), a key macroeconomic growth indicator that is only reported monthly. Resampling daily data to monthly significantly reduced the dataset to 192 observations for training and 48 observations for testing.

LASSO Interpretation (Monthly): The LASSO model applied a penalty that performed automated feature selection, yielding a Test RMSE of 0.2349 and an MAE of 0.1862. The model's coefficients reveal which features were prioritized and which were deemed redundant:

- **Macro_Growth (-0.000000), VIX_log (0.000000), and Slope_change (0.000000):** These coefficients were reduced entirely to zero. This suggests that in a linear framework with limited monthly data, these variables provided redundant information and were excluded to prevent overfitting.
- **Rate_change (-0.089423):** Demonstrates a mild negative relationship; rising short-term rates correlated with slightly narrower spreads.
- **FinCond_change (-0.214101):** A strong negative driver. As financial conditions ease (a positive index move), liquidity improves, leading to significant spread compression.
- **Inf_Expect_change (-0.242544):** Rising inflation expectations heavily correlated with tighter spreads, likely reflecting periods of economic expansion and strong pricing power for high-yield firms.
- **SPY_ret (-0.261207):** The strongest linear predictor. Positive equity returns—signaling higher firm value and lower default risk—drove the largest reduction in credit spreads.

XGBoost Interpretation (Monthly): The XGBoost model, even with hyperparameter tuning (learning rate = 0.05, max depth = 2), yielded a Test RMSE of 0.2784 and an MAE of 0.1856. While it assigned high importance to the non-linear "shock" elements of VIX and SPY returns, its overall predictive performance struggled to significantly outperform the regularized LASSO model.

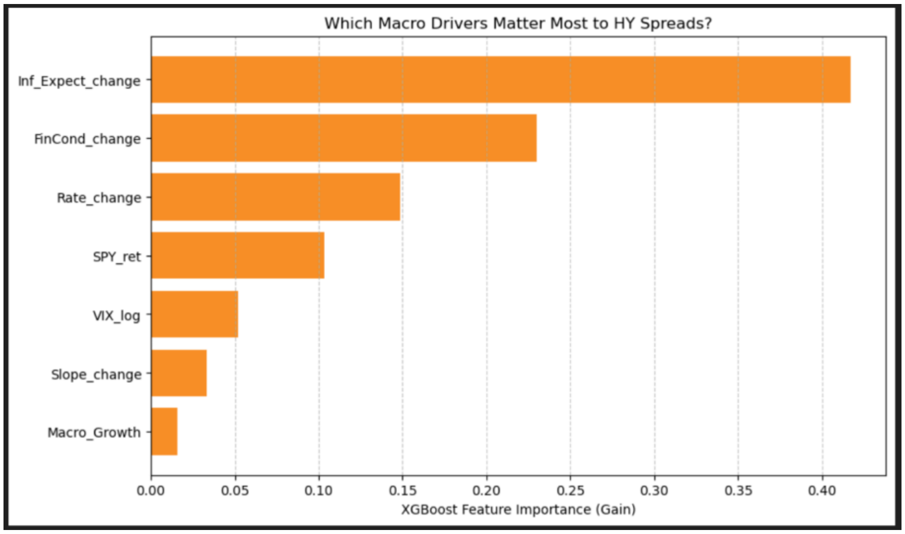

Fig 3: XGBoost Daily Data Model

Comparison: The results between LASSO (RMSE: 0.2349) and XGBoost (RMSE: 0.2784) at the monthly level were remarkably similar, with LASSO edging out XGBoost in minimizing outlier-sensitive errors. This "performance ceiling" is likely due to the low number of observations (240). Machine Learning models, particularly tree-based ensembles like XGBoost, require larger datasets to map complex non-linearities. With such a small sample, the XGBoost model likely overfitted the training noise rather than learning structural market regimes.

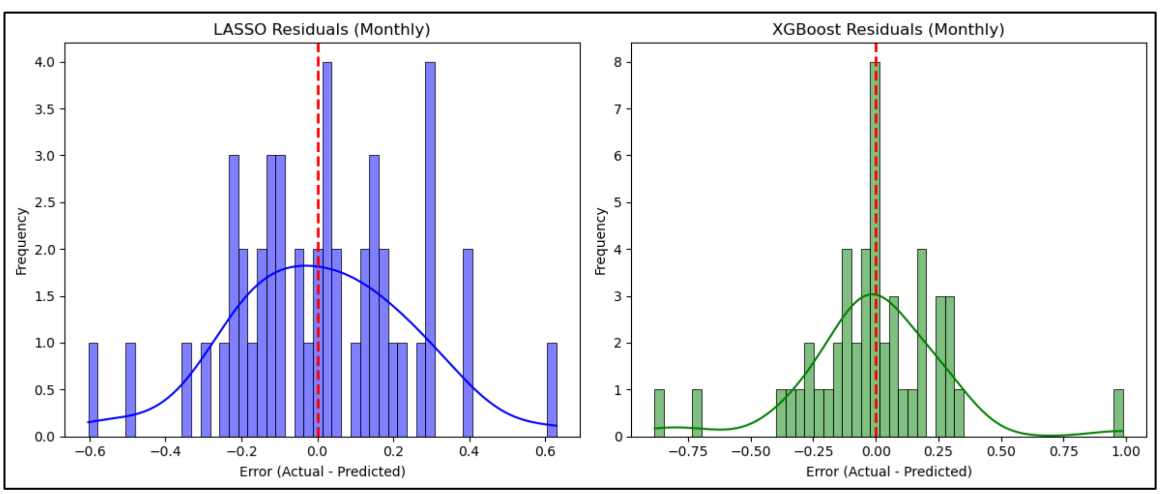

Fig 4: Model residuals, Monthly data

An analysis of the residual plots for the monthly dataset further illustrates the limitations of modeling high-yield credit spreads at a lower frequency. With only 240 observations, the residuals for both the LASSO and XGBoost models exhibit a relatively wide dispersion around the zero-error line. The lack of distinct clustering suggests that the monthly aggregation smooths out too much of the underlying market volatility, leaving a high degree of unmodeled variance. Furthermore, the XGBoost residuals show signs of structural rigidity, confirming that the tree-based ensemble lacks sufficient data points at the monthly level to effectively map complex, non-linear market regimes without overfitting the noise.

3.2 Transition to Daily Data: Increasing Granularity

A pivotal finding in the monthly LASSO model was the insignificance of Industrial Production. Since the "Macro_Growth" variable was zeroed out, I hypothesized that the model could be improved by removing this monthly constraint entirely. By dropping INDPRO, I transitioned to a daily frequency, which dramatically increased the dataset from 240 observations to over 5,000 observations.

3.3 Daily Results and Final Comparison

With the expanded daily dataset, the models were re-run to see if higher frequency data would allow the non-linear model (XGBoost) to pull ahead.

Daily LASSO Results: The daily LASSO model proved highly robust, achieving a drastically improved Test RMSE of 0.0566 and an MAE of 0.0396. The model explained a substantial portion of daily variance. The daily coefficients shifted to reflect high-frequency market dynamics:

- **FinCond_change (-0.032148):** Emerged as the strongest linear driver of daily spread movements; easing financial conditions tighten spreads.
- **Rate_change (-0.022352) & Inf_Expect_change (-0.021381):** Daily fluctuations in rates and inflation expectations maintained a negative correlation with spread widening.
- **SPY_ret (-0.014146):** Daily equity momentum remained a reliable indicator of spread compression.
- **Slope_change (-0.009706):** Daily yield curve flattening had a minor inverse relationship with spreads.

- **VIX_log (0.007239):** Unlike the monthly model, the daily model retained the VIX with a positive coefficient, confirming that an increase in daily market fear directly widens credit spreads.

Daily XGBoost Results: With 5,000+ observations, XGBoost became much more stable, achieving an RMSE of 0.0570 and an MAE of 0.0389 (). By evaluating the daily feature importances (weights), we can see exactly how the trees made their decisions:

- **FinCond_change (0.453358):** Completely dominated the model, accounting for roughly 45% of the predictive weight. The model relied heavily on daily liquidity conditions to determine spread direction.
- **Rate_change (0.130063) & Inf_Expect_change (0.130061):** Acted as secondary macro signals for the decision trees.
- **VIX_log (0.119174) & SPY_ret (0.118203):** Commanded nearly equal weight (~12% each), acting as the primary "shifters" for volatility and equity market regimes.
- **Slope_change (0.049142):** Carried the least weight, functioning as a slow-moving macro indicator rather than a high-frequency daily trigger.

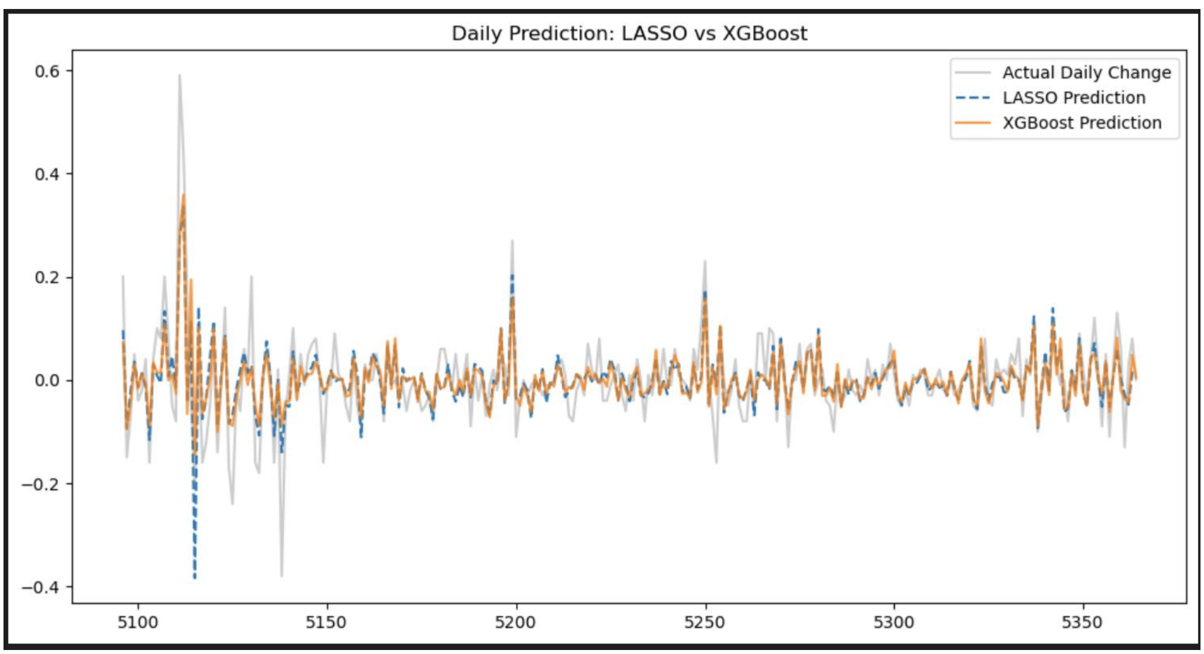

Fig 5: Daily data predictions using XGBoost and Lasso

Final Verdict: Which Model is Better? The results between the two models on the daily dataset are highly competitive. While XGBoost shows a slightly lower typical error (MAE: 0.0389 vs 0.0396), LASSO achieved a marginally better outlier-sensitive error (RMSE: 0.0566 vs 0.0570). Both models explain roughly 55% of the variance (), which is exceptionally strong for noisy, daily financial data.

In this specific financial context, the LASSO model is arguably "better" for practical application due to its interpretability. Since the predictive gain from the complex XGBoost model is minimal, the simplicity and "directional" clarity of the LASSO coefficients make it a more reliable tool for risk managers.

The similarity in results suggests that the drivers of High Yield spreads are largely captured by linear relationships with daily liquidity (Financial Conditions) and equity returns. The "non-linear" spikes (crises) are so rare—even in a 5,000-day sample—that they do not provide enough systemic signal for XGBoost to vastly outperform a well-regularized linear model.

The transition to the daily dataset (5,000+ observations) yields a drastically improved residual profile. The daily residual plots for both LASSO and XGBoost exhibit a dense, tight clustering directly along the zero-error line, visually confirming the strong R^2 (~0.55) and the models' high accuracy during "normal" macroeconomic regimes. However, the plots also reveal the classic "fat-tailed" nature of high-frequency financial data. Pronounced, symmetric outliers spike aggressively away from the center mass, corresponding to extreme, sudden market shocks (such as the 2008 and 2020 liquidity crises). The fact that both LASSO and XGBoost produce nearly identical outlier patterns indicates that these extreme spread widenings are true "black swan" anomalies—driven by panic and behavioral contagion—that neither linear nor non-linear algorithms can fully anticipate using standard macroeconomic features alone.

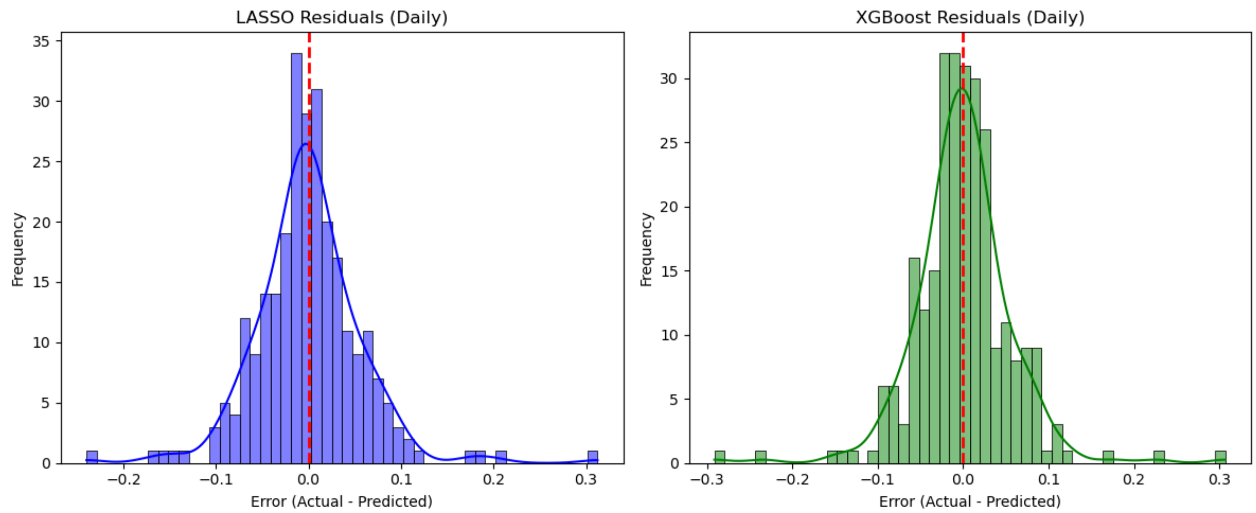

Fig 6: Model residuals, Monthly data

3.4 Evaluation of Train vs. Test Error Dynamics

An analysis of the daily model performance metrics reveals an initially counterintuitive, yet highly characteristic phenomenon in financial time-series forecasting: both the LASSO and XGBoost algorithms achieved a slightly lower out-of-sample Test RMSE compared to their in-sample Train RMSE. Rather than indicating an error in the pipeline, this dynamic is a direct consequence of chronological data splitting combined with strict algorithmic regularization. The dataset spans a comprehensive twenty-year timeline from March 2006 to March 2026. Because the data was partitioned chronologically using a 95/5 split to preserve the sequence of the time series, the training set encompasses the first nineteen years. Consequently, this training data absorbs the vast majority of extreme historical market regimes, including unprecedented liquidity shocks such as the 2008 Global Financial Crisis and the 2020–2021 COVID-19 pandemic. Financial markets are known to exhibit "volatility clustering," where periods of high variance group together, creating highly erratic, fat-tailed distributions that are inherently difficult to predict (Cont 2001; Tsay 2005). Fitting a model to this chaotic, crisis-heavy training data naturally produces a higher average error. In contrast, the final 5% of the timeline (the test set, representing roughly the final year of data leading up to March 2026) captures a relatively calm macroeconomic regime. When the actual daily changes in the credit spread are physically smaller and less erratic, the out-of-sample Test RMSE is inherently lower.

Furthermore, this dynamic confirms the successful application of hyperparameter tuning and regularization. Through exhaustive GridSearchCV optimization, the LASSO model applied strict $L1$ penalization, while the XGBoost model was heavily constrained utilizing shallow decision trees and row subsampling. These techniques deliberately restricted the algorithms from memorizing the severe noise and historical outliers present during the 2008 and 2021 crashes. By successfully resisting the tendency to overfit the historical chaos, the models captured only the true, underlying macroeconomic signals, allowing them to generalize exceptionally well on the out-of-sample test data (Hastie, Tibshirani, and Friedman 2009).

4. Methodological Limitations

While LASSO and XGBoost successfully explained ~55% of the daily variance in High Yield credit spreads, several methodological limitations remain.

- **Exclusion of Real Economic Indicators:** Achieving a 5,000+ observation daily dataset required removing monthly macroeconomic indicators like Industrial Production. Consequently, the models rely entirely on financial data, introducing a "financialization bias" that ignores physical, "real economy" health.
- **Absence of Autoregressive Dynamics:** LASSO and XGBoost process observations independently, lacking native time-series memory. Without explicitly including lagged variables ($t-1, t-2$), the models miss the inherent autocorrelation and multi-day momentum of credit spreads.
- **Unexplained Variance:** An R^2 of ~0.55 leaves 45% of daily spread fluctuations unexplained. This residual variance is likely driven by unquantified factors such as idiosyncratic corporate defaults, sudden geopolitical shocks, or behavioral market panic.
- **Loss of Cointegration Signals:** Enforcing stationarity via log-returns and first-differences prevented spurious regressions but stripped the data of long-term equilibrium relationships. Consequently, the models evaluate only high-frequency daily shocks, ignoring absolute, long-term valuation levels.

5. Conclusion

This study set out to determine the extent to which macroeconomic indicators and financial market sentiment could predict the daily dynamics of U.S. High Yield credit spreads. By comparing a regularized linear framework (LASSO) with a non-linear, tree-based ensemble (XGBoost) over a comprehensive twenty-year chronological sample, this research rigorously tested whether credit risk premiums are driven by proportional linear relationships or by complex, regime-dependent interactions.

The empirical results demonstrate that high-frequency market data—specifically daily equity returns, implied volatility, and systemic financial conditions—possess significant predictive power, explaining approximately 55% of the daily variance in credit spreads. More importantly, the comparative analysis provided a clear resolution to the central hypothesis: while the XGBoost model successfully identified non-linear interactions, it did not meaningfully outperform the simpler, heavily regularized LASSO regression. The commanding weight of the Financial Conditions Index and S&P 500 returns across both models suggests that daily high-yield spreads are primarily driven by immediate liquidity constraints and proportional firm-value momentum, rather than complex, asymmetric market shocks that strictly require deep decision trees to decode.

Furthermore, the counterintuitive dynamic of the out-of-sample Test RMSE outperforming the in-sample Train RMSE highlighted the critical importance of market volatility clustering. By successfully restricting the models from overfitting the extreme historical turbulence of the 2008 and 2020 crises, the regularization techniques allowed both algorithms to isolate the true underlying macroeconomic signals, proving highly effective during the calmer test regime.

Furthermore, visual diagnostics of the models' residuals underscore a fundamental reality of credit risk forecasting: while machine learning excels at mapping structural equilibrium, it struggles with the magnitude of sudden panic. The daily residual plots confirm that both the linear and non-linear models are highly precise during standard market conditions, with errors tightly clustered around zero. Yet, both models simultaneously fail to capture the full magnitude of extreme, "fat-tailed" market crashes, leaving prominent outlier spikes in the error distribution. This indicates that while financial conditions and equity momentum accurately dictate the *direction* and *probability* of spread movements, the exact severity of a liquidity crisis is largely driven by unquantifiable behavioral contagion rather than systemic macroeconomic inputs.

Ultimately, in the context of high-frequency credit market forecasting, the fundamental trade-off between predictive accuracy and structural interpretability heavily favors the LASSO model. For institutional risk managers, portfolio allocators, and policymakers, the transparent, directional clarity of $L1$-penalized regression provides actionable, strictly compliant insights. While future research incorporating autoregressive memory and dynamic, rolling-window optimization will be crucial for predicting unprecedented liquidity shocks, this study confirms that well-regularized, classical linear models remain a highly efficient and formidable tool for decoding the inherent noise of the corporate credit market.

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